ME5412 Robotics for Healthcare

Text to robot signing

Finite expressions and a learned trajectory prototype



Simulation prototype / 8 October 2026

Candidate motion. ASL comprehensibility unreviewed.

Problem and project scope

Proposed setting

Brief service interactions at a care reception desk.

The project tests expression selection and robot motion. It has no clinical outcome evidence.

PLEASE_WAIT currently approximates the WAIT word example.

Design alternatives

Text display: no robot motion
Human signing: reference for meaning
Robot candidate: repeatable motion to inspect

Current vocabulary

HELLO
THANK_YOU
PLEASE_WAIT

Default demonstration architecture

01	**English input or phrase button**	One complete meaning
02	**Shared guard and selector**	execute / clarify / unsupported / error
03	**Action ID contract**	HELLO / THANK_YOU / PLEASE_WAIT
04	**Independent preview permission**	Unreviewed motion blocked by default
05	**MuJoCo trajectory and video**	Named joints and timestamped positions

The selector returns an ID. It never generates joint commands.

Robot model and control boundary

Unitree G1 + Inspire DFQ

53 scalar joint coordinates
12 retained mimic relations
0 actuators in this scene



Fixed base with kinematic qpos replay.
Coupled fingers limit independent handshape control.

Dynamics and ASL comprehension remain untested.

Video references guide qualitative calibration



Near lower-face proxy



Forward and slightly down



Final palm faces up

ASLU video observations: 0.916 s, 1.116 s and 1.516 s

Human video provides qualitative references. Robot keyframes use manual engineering targets.

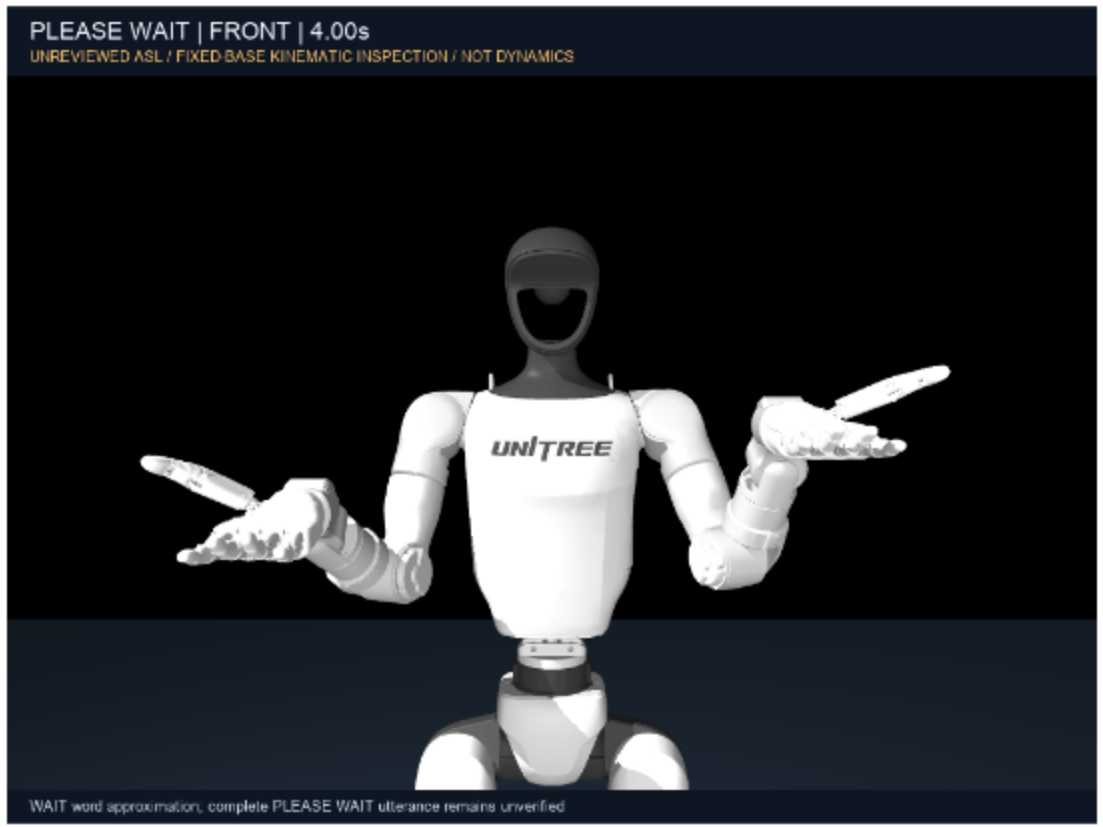
Motion construction and checks

Manual wrist targets

Bounded inverse kinematics fits position and orientation while limiting posture change.

$$q(u) = (1 - s)\, q_0 + s\, q_1$$
$$s(u) = 3u^2 - 2u^3$$

u is normalized segment time.
Zero endpoint velocity does not ensure continuous acceleration.



WAIT remains a coupled-finger approximation.

Three selectors share one interface

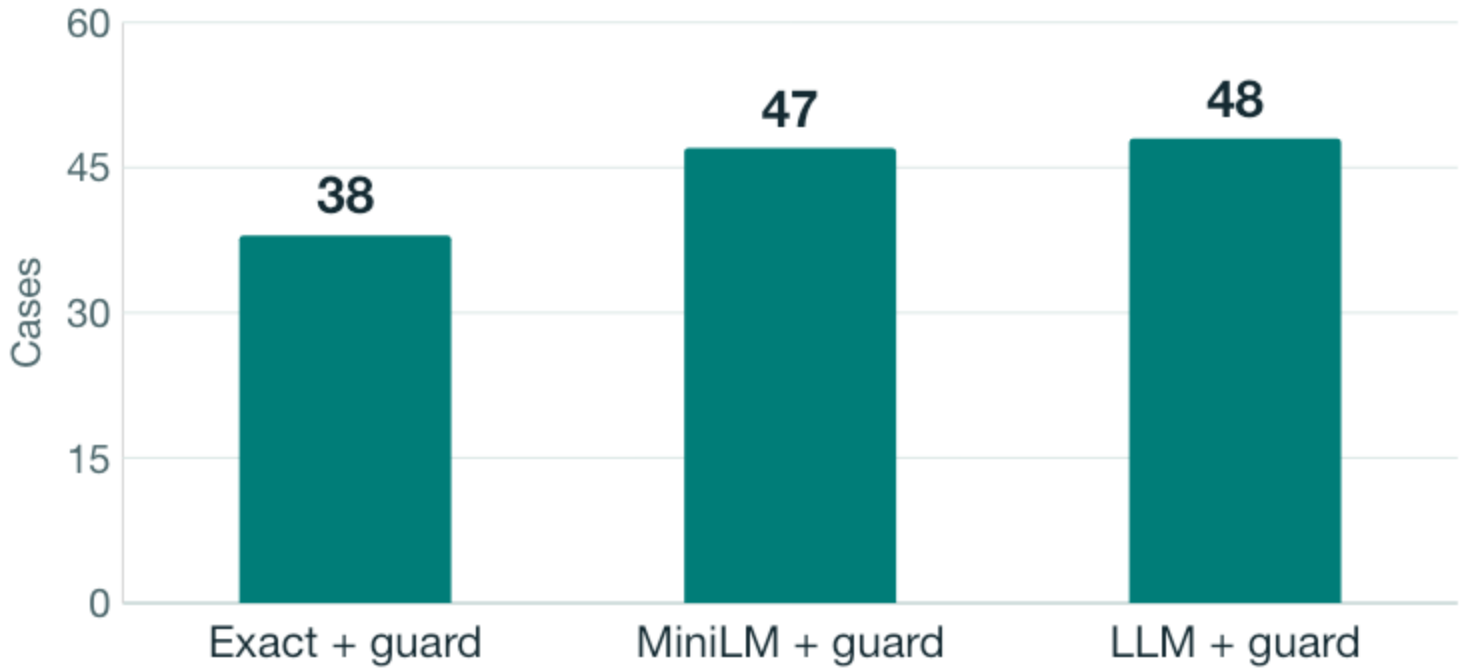
Method	Mechanism	Main limitation
Exact matching	Full registered phrase after shared guard	Limited paraphrase coverage
MiniLM	Local 384-dimensional sentence embeddings	Thresholds can reject valid paraphrases
Local LLM	Strict five-label output from existing Q8 checkpoint	Wrong choices and invalid labels observed

All methods return the same allowed ID or a non-execution status.

Default demo: exact or MiniLM. LLM remains disabled after its development gate failed.

Frozen text evaluation

60 cases: 30 supported and 30 out of scope



18 development cases

Test frozen before evaluation

Agent-drafted labels
Human review pending

Correct behavior includes correct rejection. This metric does not measure ASL comprehension.

The LLM score includes incorrect execution

Outcome	Exact	MiniLM	Local LLM
Supported input selected correctly	8 / 30	17 / 30	20 / 30
Wrong ID on supported input	0 / 30	0 / 30	1 / 30
Execution on out-of-scope input	0 / 30	0 / 30	2 / 30
Invalid output / runtime error	0 / 60	0 / 60	2 / 60

35 actual LLM calls

25 cases ended in the shared guard

The LLM remains disabled by default. Small-sample zero counts do not establish zero risk.

Demonstration and failure behavior

1 “Thank you.”

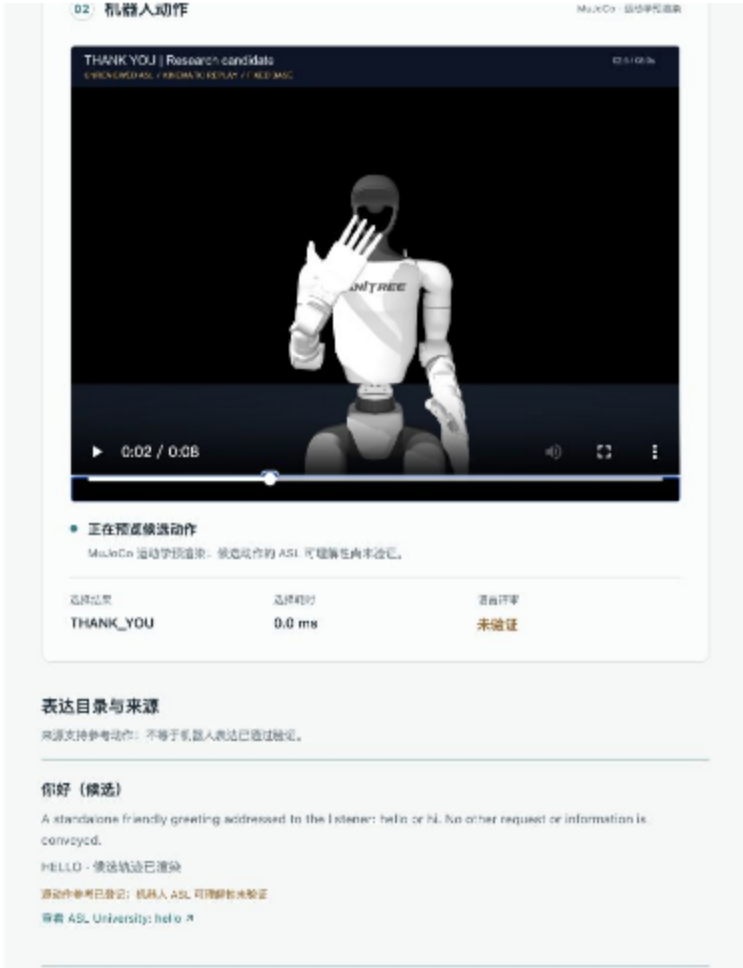
Exact match selects THANK_YOU. Preview stays blocked.

2 Explicit candidate preview

Research permission enables the pre-rendered motion.

3 A mixed or negated request

The system rejects the input and clears the old video.



Local startup: run_demo.command

Learning trajectories from text and phase

Frozen MiniLM (384) + Gaussian phase RBFs (40)

Ridge regression fits 26 active joint coordinates.
Limit projection and mimic reconstruction return 53 coordinates.

Candidate	Odd-frame RMSE	Max error
HELLO	0.000430 rad	0.004278 rad
THANK_YOU	0.002528 rad	0.046329 rad
PLEASE_WAIT	0.000621 rad	0.005549 rad

3 authored demonstrations

367 even frames for fitting
365 odd frames for validation

Same-demonstration interpolation only

Fit and sampled geometry gates passed. Default route unchanged. ASL review pending.

How2Sign text records and missing motion targets

35,263 metadata rows acquired and hashed

Official split	Annotation rows
Train	31,165
Validation	1,741
Test	2,357
Total	35,263

Available locally

English text, sentence IDs and aligned timestamps

New video / pose / robot supervision: 0

Next step: permitted motion targets, robot retargeting and independent demonstration splits.

Six owners and remaining work

Owner	Current artifact	Next acceptance check
A1 Sources	Video notes and text index	Permitted motion targets and meaning review
A2 Motion	Trajectories and fitted RBF model	Independent demonstrations and retargeting
B1 Matching	Exact and local embeddings	New reviewed development data
B2 LLM	Reproducible negative result	New protocol before prompt changes
C1 Evaluation	Frozen cases and raw outputs	Independent labels and new test set
C2 Integration	Local demo and report	Second-computer reproduction

26–30 Oct: rehearsal materials. 31 Oct–2 Nov: final rehearsal. 3 Nov: slides due.

Conclusions and open questions

Working engineering prototype

The default selector and candidate motion pipeline remain available.
A fitted text–phase model now reproduces three authored trajectories.

Open language question

Three demonstrations support temporal interpolation evidence.
New-sign generalization and ASL comprehension remain untested.

Next evidence: real motion targets, independent demonstrations and meaning assessment.